

Understanding Modular Building Risk and Opportunities



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Agenda

Modular 101

Understanding market for modular building, growth drivers, and quality phases

- Expanding market
- Benefits
 - Project performance
 - Environment impact
 - Worker Safety
- Quality phase overview

Panel Q&A

Insights into **quality issues** stemming from modular, **emerging trends** and considerations for risk managers, and **insurance coverage** issues



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Modular 101

Understanding market for modular building, growth drivers, and quality phases



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Permanent Modular Construction (PMC) Reached \$12B in 2022 North America

\$12B

2022 Starts

Modular accounts for 6% of 2022 construction starts in North America

86M

2022
Modular sq. ft.

Modular Building Institute estimate of modular output

6%

2022
Market Share

Modular market has doubled in 5 years

1.4B

2022
Total Sq. ft

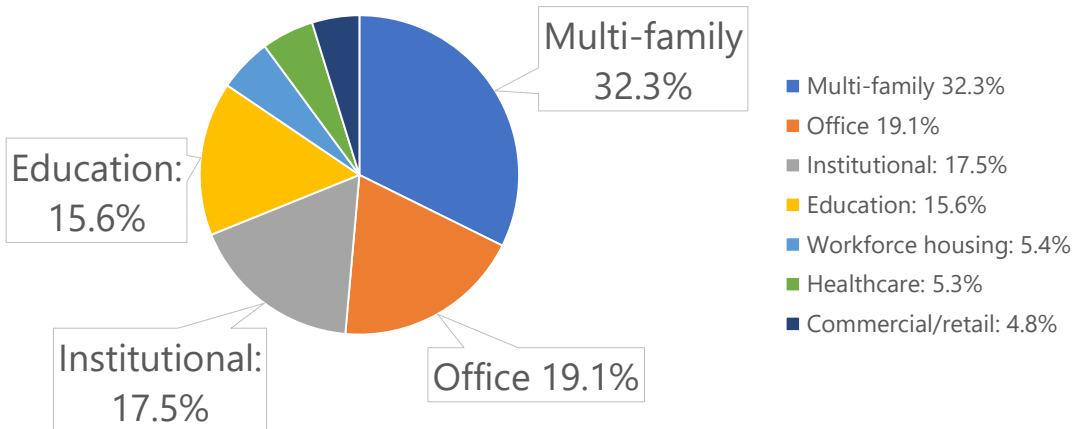
Construct Connect reported new, non-residential construction put in place

Source: www.modular.org Modular Building Institute 2023 Permanent Modular Construction Report



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PMC Becoming Prevalent in Non-Residential Construction Sectors



Source: www.modular.org Modular Building Institute 2023 Permanent Modular Construction Report



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What's Driving the Growth in PMC Modular Building?

Construction costs & uncertainty

~55-60% of all construction projects run over schedule and over budget

Using pre-fab/modular inputs:

91% of general contractors reported better budget performance

48% observed >10% cost savings

Construction Schedule

- ✓ Modular homes construction 30-50% faster
- ✓ Onsite productivity - integrating modular component

Materials

- ✓ Costs more predictable through bulk ordering direct from manufacturer
- ✓ Offsite storage reduces typical weather-related delays or impediments

Labor

- ✓ Offsite construction is done via assembly lines and automation
- ✓ Production efficiency
- ✓ Requires fewer specialized skills than typical construction - subcontractor labor onsite costly

Source: www.modular.org Modular Building Institute 2023 Permanent Modular Construction Report



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Modular Supports Sustainable Business, ESG Initiatives, May Ease Labor Challenges



Environmental Impact

- Reduce waste – less rework
- Reduce carbon footprint
- Can relocate entire buildings or repurpose modules
- Energy efficiency and tight building envelope



Worker Safety

- Broaden available labor pool
- Offsite construction reduces onsite hazards
- Safer workplaces may lead to greater employee longevity and higher productivity

Source: www.modular.org Modular Building Institute 2023 Permanent Modular Construction Report



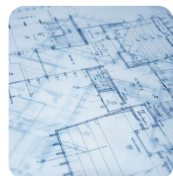
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Modular Production & Quality Assurance Phases

Manufacturing Facility



Industrialized building: construction, design, and manufacture factory



Design approved by end-user and regulating authorities



Fabricate modular components in controlled factory



Transport modules to the final building destination



Integrate modular units, finish building site

Each phase introduces logistical, regulatory, contractual, and insurability concerns

Source: www.modular.org Modular Building Institute 2023 Permanent Modular Construction Report



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Modular 2023 Panel Q&A

Insights into **quality issues** stemming from modular, **emerging trends** and **considerations** for risk managers, **insurance coverage** issues



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Do You See Increasing Innovative Tech Interactions in Modular Projects?



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Virtual Design
Construction



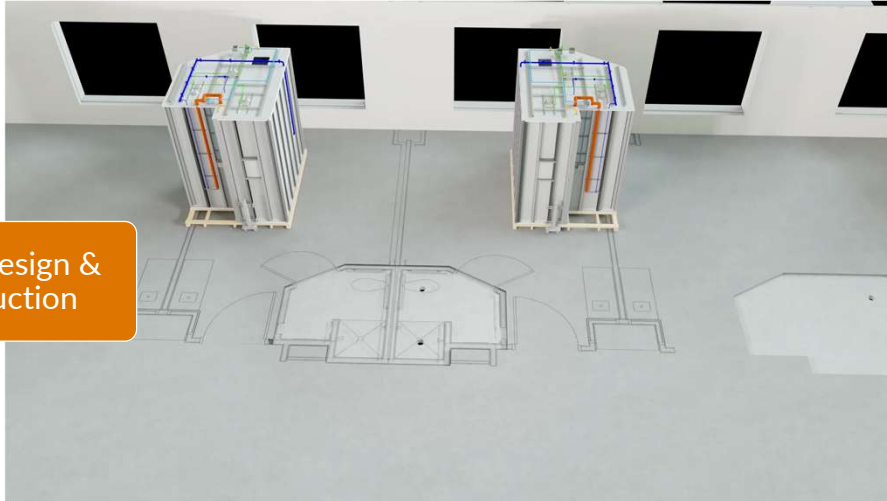
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Automation & Robotics
Technology



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Technology is used throughout the modular design and installation phases, starting with the model



Virtual Design & Construction

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Technology used in the plant environment

Robotic welder



Computer Numerical Control Machine



Numbered Framing Members



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Technology interactions at the project site

Panel installation /
lift



Laser Scanning



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How Could Modular Change the Risk?



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Jurisdictions

Licensing
Insurance

Contractual Risk

Supply agreements
Temporary Labor

Transportation Risks

Quality - onsite
Insuring manufacturing
facility



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What Should Insurance Managers Consider for Coverage Needs on Modular Projects?



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Offsite Manufacturing
Facility

CGL Scope &
Coverage Impact

Insurance Program
Nuances



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Effective Risk Management Rooted in Awareness of How Insurance Responds

- Defective design
- Defective manufacturing
- Defective installation
- Damage during transportation

What risks are
common?

What coverage /
policy could respond?

- Liability & Professional
 - General Contractor
 - Subcontractor/Installer
 - Manufacturing company
- Builders Risk (BR)
- Subcontractor Default Insurance (SDI)
- Surety



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How General Liability Coverage May Respond to Property Damage During Modular Project

General Liability Premises or Operations Hazards: damage may stem from 4 key categories

Damage to prefabricated modules

Damage to your work/on your behalf

Damage to that particular part of others' work

Delays from design, causing Bodily Injury/Property Damage

CGL
Exclusion j.
Damage to
Property

(4) Personal property in the care, custody or control of the insured;

(5) That particular part of real property on which you or any contractors or subcontractors working directly or indirectly on your behalf are performing operations, if the "property damage" arises out of those operations; or

(6) That particular part of any property that must be restored, repaired or replaced because "your work" was incorrectly performed on it.

Source: Insurance Services Office, Inc © 2012. COMMERCIAL GENERAL LIABILITY COVERAGE

Subject to any professional or errors & omissions exclusions



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How General Liability Coverage May Respond to Property Damage Occurring Post-Project

General Liability Products or Completed Operations Hazards: Damage may stem from 3 key categories

Damage to your product

- Your product includes goods you manufacture, sell, handle, distribute or dispose of

Damage to your work

- Your work includes operations you perform, or performed on your behalf
- AND materials, parts, equipment

Damage to your subcontractors' work

k. Damage To Your Product

"Property damage" to "your product" arising out of it or any part of it.

l. Damage To Your Work

"Property damage" to "your work" arising out of it or any part of it and included in the "products-completed operations hazard".

This exclusion does not apply if the damaged work or the work out of which the damage arises was performed on your behalf by a subcontractor.

Source: Insurance Services Office, Inc © 2012. COMMERCIAL GENERAL LIABILITY COVERAGE FORM CG 00 01 04 13



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Project-Specific, Controlled Insurance Programs Have Nuances...

1. Scope of CIP

- **Controlled Insurance Programs** typically only cover well defined project sites
- **Schedule Off-site locations** such fabrication facilities & staging areas to effect coverage
- **Clarify scope** with your carrier if components manufactured off-site are covered by the CIP

2. Transportation

- Obtain coverage during transit since this will likely NOT be covered by the wrap CGL policy
- Note complexity if components are pre-fabricated off-shore and shipped to the U.S.

3. Products Liability

- If coverage for fabricated components need to be found outside the CIP, a wrap exclusion under the subcontractors' policies will likely apply
- Contractor fabricating modular-if the entity goes out of business, coverage gap is likely in products or completed operations coverage



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Do Construction Firms Have Greater Design Liability for Modular Build? What Coverage Enhancements Do You Recommend They Consider?



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Professional liability



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Contractors E&O

Manufacturer E&O



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What Contractual Issues Present? Has the Industry Responded with Improved Contracts?



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Modular design

Manufacturing

Construction



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Have
Questions?



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